

Total No. of Questions : 8]

SEAT No. :

PB-4949

[Total No. of Pages : 2

[6263]-622

B.E. (Computer)

EMBEDDED AND REAL TIME OPERATING SYSTEMS
(2015 Pattern) (Semester - VIII) (410252C)

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidates:

- 1) *Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.*
- 2) *Neat diagrams must be drawn wherever necessary.*
- 3) *Figures to the right side indicate full marks.*
- 4) *Assume Suitable data if necessary.*

- Q1)** a) Explain Performance metrics for processor of embedded system? [7]
b) What is the use of ROM image in Embedded system? Draw a structure of ROM image and explain various components embedded inside it [7]
c) What are types of serial communication in embedded systems? [6]

OR

- Q2)** a) Explain timing diagram of Analog to Digital Converter (ADC)? [7]
b) Explain block diagram of Embedded systems? [7]
c) Explain ZigBee protocol in detail? [6]

- Q3)** a) Explain Hard versus soft real-time systems and their timing constraints? [9]
b) What are Functional parameters and Resource parameters of real-time process? [8]

OR

- Q4)** a) Explain Real-time task and task states with example? [9]
b) What are Approaches to real-time scheduling? [8]

P.T.O.

Q5) a) How to Enforcing mutual exclusion and critical sections in embedded systems? **[9]**

b) What are Effects of resource contention and resource access control?**[8]**

OR

Q6) a) Explain Inter-process communication in RTOS? **[9]**

b) What are Interrupt routines in RTOS environment? **[8]**

Q7) a) Explain synchronization in Real-time Operating system? **[8]**

b) What are different Medium access-control protocols for broadcast networks? **[8]**

OR

Q8) a) Explain Software development process for embedded system? **[8]**

b) What are Embedded software development tools/ **[8]**
